ICT 4442 - Deep Learning Mini Project

Phase 2: Interim Report

Title: Comparative Analysis of Deep Learning Architectures for Sentiment Analysis of Movie Reviews

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Repository: https://github.com/HarshavardhanReddy585/imdb-sentiment-analysis-ict4442

Submission date: 02 October 2026 (late submission; seek faculty approval for acceptance)

1. Literature Review

Paper	Method	Dataset	Key result	Relevance
Maas et al. (2011) [1]	Unsupervised word vectors + sentiment classifier	IMDb 50K	Established a large labelled/unlabelled movie-review benchmark and showed domain-aware representations help sentiment classification.	Defines the dataset and benchmark used here.
Kim (2014) [2]	1D CNN with multiple filter widths	Sentence classification benchmarks	Simple CNN filters capture informative local phrases and perform strongly with limited task-specific architecture.	Supports the project CNN architecture and max-pooling design.
Hochreiter & Schmidhuber (1997) [3]	Long Short-Term Memory recurrent units	Sequence modelling	Gated memory addresses vanishing-gradient limitations in recurrent learning.	Provides the basis for the planned BiLSTM model.
Tai, Socher & Manning (2015) [4]	Tree-LSTM and compositional recurrent encoders	Sentiment Treebank	Structured recurrent encoders improve representation of compositional meaning.	Motivates context-aware recurrent comparison and error analysis.
Mikolov et al. (2013) [5]	Word2Vec distributed word representations	Large text corpora	Efficient embeddings encode useful lexical regularities.	Informs the embedding representation used by non-Transformer models.
Peters et al. (2018) [6]	ELMo contextual representations	Multiple NLP tasks	Deep contextual representations transfer across tasks and improve over static embeddings.	Connects embedding-average baselines to contextual pretraining.
Vaswani et al. (2017) [7]	Transformer self-attention	Translation and sequence tasks	Self-attention models token-to-token dependencies without recurrence.	The architectural foundation for the planned DistilBERT comparison.
Devlin et al. (2019) [8]	BERT bidirectional pretraining + fine-tuning	GLUE, SQuAD and other tasks	Bidirectional masked-language pretraining produced state-of-the-art results across language tasks.	Establishes the pretrained attention-family reference point.
Sanh et al. (2019) [9]	DistilBERT knowledge distillation	General language understanding	A smaller model retains most BERT capability with lower size and faster inference.	Specifies the fourth proposed model and its efficiency trade-off.
Howard & Ruder (2018) [10]	Universal Language Model Fine-tuning	Text classification benchmarks	Transfer learning and discriminative fine-tuning improve text classification.	Supports pretrained-model baselines and controlled fine-tuning.

The literature indicates a progression from static or pooled word representations, through local convolution and recurrent context, to bidirectional self-attention with pretraining. The interim comparison therefore keeps the data split and metrics fixed while varying the architecture family.

2. Dataset Acquisition & Preprocessing

The project uses the IMDb Large Movie Review Dataset [1], consisting of 25,000 labelled training reviews and 25,000 labelled test reviews, balanced between positive and negative classes. The official archive is downloaded locally under `data/` when running the pipeline and is excluded from Git.

The reproducible pipelines are implemented in `src/train_scratch_models.py` and `src/train_bilstm.py`:

1. Read review text from the official `train/neg`, `train/pos`, `test/neg`, and `test/pos` folders.

2. Lowercase text and tokenize alphabetic words plus apostrophes; retain sentiment-bearing words such as `not`.

3. Shuffle the official training split with seed 42 and divide it into 80% train and 20% validation. The official test split remains untouched until final evaluation.

4. Construct the vocabulary from the training partition only; keep the 12,000 most frequent tokens plus <pad> and <unk>.

5. Encode each review as a fixed-length sequence, padding shorter reviews and truncating longer reviews deterministically. The MLP/CNN smoke runs use 512 tokens; the BiLSTM smoke run uses 40 tokens to keep CPU runtime manageable.

6. Use the same encoded partitions and metrics for every model. No test-set score is used for model selection.

The implementation is deliberately CPU-friendly for the delayed interim submission. It supports a small reproducibility run through `--max-train` and `--max-test`; the final report must rerun the same protocol at full scale and include BiLSTM and DistilBERT.

3. Models Implemented So Far

Model	Owner	Status	Preliminary metric	Notes
MLP with averaged learned embeddings	Kshitij Singh	Implemented in `src/models/mlp.py` and `src/train_scratch_models.py`	Validation accuracy 75.75%, F1 0.753	Two-epoch PyTorch interim run on 3,200 train / 800 validation reviews with 512-token inputs; test accuracy 69.2%, F1 0.686.
1D-CNN with max-pooled multi-width filters	Ratul Jha	Implemented in `src/models/cnn1d.py` and `src/train_scratch_models.py`	Validation accuracy 61.5%, F1 0.693	Two-epoch PyTorch interim run on the same split with 512-token inputs; test accuracy 58.2%, F1 0.671 and a positive-class recall bias remains.
BiLSTM	Harshavardhan Reddy	Implemented in `src/models/bilstm.py` and `src/train_bilstm.py`	Validation accuracy 46.5%, F1 0.389; smoke test accuracy 51.0%, F1 0.372	Two-epoch CPU smoke run on the common 800/200/200 split; needs tuning and full-scale rerun.
DistilBERT	Meesha Gupta	Planned / not evidenced in current folder	Pending	Fine-tune after baseline comparison; record model version and hyperparameters.

The reported values come from real runs with seed 42 and the official IMDb folder split. MLP/CNN use 3,200 training, 800 validation, and 1,000 test reviews with 512-token inputs; BiLSTM uses a smaller 800/200/200 CPU smoke split with 40-token inputs. They are preliminary and intentionally not presented as final performance. The generated JSON files are checked into the repository as evidence.

4. Individual Contribution Log

4A. Completion Evidence Available at Interim Submission

The following evidence is available in the canonical GitHub repository at the time of this draft:

- Official IMDb archive downloaded locally and excluded from Git through `.gitignore`.
- Reproducible preprocessing and training scripts under `src/`.
- MLP and 1D-CNN PyTorch implementations with saved validation/test metrics under `results/`.
- BiLSTM implementation with a CPU smoke-run metric file under `results/`.
- Two commits documenting the model/evidence work: `8544666` and `6dee82d`.
- Python compilation and model-import checks completed successfully.

The MLP and CNN results are interim evidence from a 3,200-train / 800-validation / 1,000-test subset. They are not claimed as final full-dataset scores. DistilBERT, full-scale four-model training, learning curves, and systematic error analysis remain outstanding.

Member	Registration no.	Task completed to date	Signature
Meesha Gupta	230953090	DistilBERT model ownership; literature/repository evidence to be added by member	_____
Ratul Jha	230953338	1D-CNN model ownership; preprocessing/model review to be verified by member	_____
Harshavardhan Reddy	230953396	Report integration, preprocessing pipeline, experiment orchestration	_____
Kshitij Singh	230953460	MLP model ownership; evaluation and comparison to be verified by member	_____

Git evidence: The canonical private repository is linked above. The owner-specific files and smoke metrics are present, but member attribution and signatures must still be verified honestly. Do not simulate signatures or claim work that a member did not perform.

5. Risk / Plan for Remaining Work

Risk	Mitigation	Target
Late Phase 2 submission may require faculty approval.	Submit immediately with a concise late-submission explanation and retain the complete evidence package.	02 Oct 2026
CPU/runtime constraints for full IMDb experiments.	Run smoke test first, then full training in stages; checkpoint metrics and record hardware/runtime.	03-05 Oct
BiLSTM and DistilBERT not yet evidenced locally.	Add both models to the shared preprocessing/evaluation interface; assign one owner per model.	06-10 Oct
Inconsistent comparisons across models.	Freeze split, vocabulary/tokenizer policy, metrics, seed, and held-out test procedure.	10 Oct
Missing contribution verification.	Push member-specific commits and obtain signed Part B/C statements.	12 Oct
Error analysis and final figures delayed.	Save confusion matrices, learning curves, and representative false positives/negatives for all models.	18 Oct

Final timeline: full four-model evaluation by 14 Oct; error analysis by 18 Oct; integrated Part C report by 27 Oct; slides/rehearsal and final code push by 31 Oct 2026.

References

[1] A. L. Maas et al., “Learning Word Vectors for Sentiment Analysis,” ACL-HLT, 2011.

[2] Y. Kim, “Convolutional Neural Networks for Sentence Classification,” EMNLP, 2014.

[3] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” Neural Computation, 1997.

[4] K. S. Tai, R. Socher, and C. D. Manning, “Improved Semantic Representations From Tree-Structured Long Short-Term Memory Networks,” ACL, 2015.

[5] T. Mikolov et al., “Efficient Estimation of Word Representations in Vector Space,” arXiv:1301.3781, 2013.

[6] M. E. Peters et al., “Deep Contextualized Word Representations,” NAACL, 2018.

[7] A. Vaswani et al., “Attention Is All You Need,” NeurIPS, 2017.

[8] J. Devlin et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” NAACL-HLT, 2019.

[9] V. Sanh et al., “DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter,” NeurIPS EMC2 Workshop, 2019.

[10] J. Howard and S. Ruder, “Universal Language Model Fine-tuning for Text Classification,” ACL, 2018.